



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2025– 2026(Odd Semester)

Degree, Semester & Branch: B.E – V Semester CSE

Code & Title: CS3551 & Distributed Computing

Name of the Faculty member (s): Mrs. P. Devisri, AP/CSE

- **Unit / Topic:** Unit 2 / Snapshot Algorithm
- **Topic Learning Outcome:** 2a
- **Activity Chosen:** Crossword puzzle
- **Justification:**
 - The **Snapshot Algorithm** is a fundamental concept in distributed systems, allowing processes to capture a consistent global state in a FIFO channel network.
 - Understanding how processes record states and channel messages is crucial for analyzing system behavior, detecting deadlocks, and debugging distributed applications.
 - A hands-on activity where students simulate the Snapshot Algorithm helps them visualize message passing, process coordination, and state recording.
 - Active participation in such real-time exercises enhances comprehension, engagement, and application skills in distributed systems.

Time Allotted for the Activity: 30 Minutes

- **Details of the Implementation:**
 - Faculty provided an overview of the Snapshot Algorithm and FIFO channels, highlighting their significance in distributed systems and how they help capture consistent global states.
 - Students were given time to individually reflect on the concept of snapshot recording, marker sending, and message tracking in FIFO channels.
 - Each student was provided with a paper containing a scenario of processes and FIFO channels.
 - Students were instructed to simulate message passing and marker sending on paper, recording the local states of each process and messages in the channels.
 - Students were encouraged to clarify doubts with the faculty while performing the simulation.
 - After completing the activity, the faculty reviewed the students' sheets, discussed the correct sequence of marker sending, state recording, and channel snapshots, and clarified common mistakes, as shown in Fig 1 and Fig 2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO10	PSO1
CO2	2	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO10	PSO1
Justification for correlation	Students will be able to understand the concept of the Snapshot Algorithm and FIFO channels in distributed systems	Students will be able to apply the algorithm correctly to record process states and messages in FIFO channels.	Students will be able to visually represent the state of processes and channels by simulating message passing and marker sending on paper.	Students' communication skills will be improved as they clarify doubts with the faculty and discuss approaches during the activity.	The Snapshot Algorithm simulation can be used to analyze and understand consistent global states, helping students apply theoretical concepts to practical distributed system scenarios.

- **Images / Screenshot of the practice:**

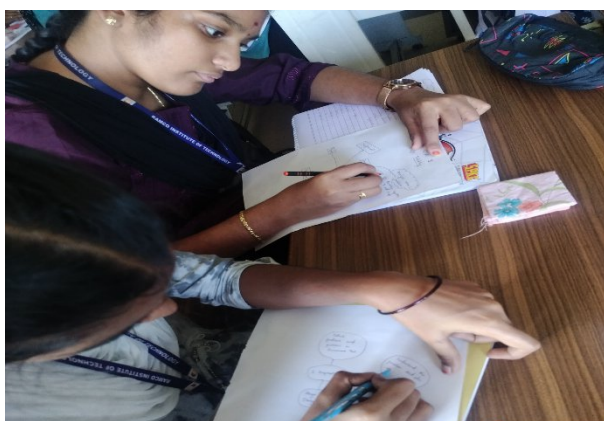


Fig 1: Students solving crossword puzzle

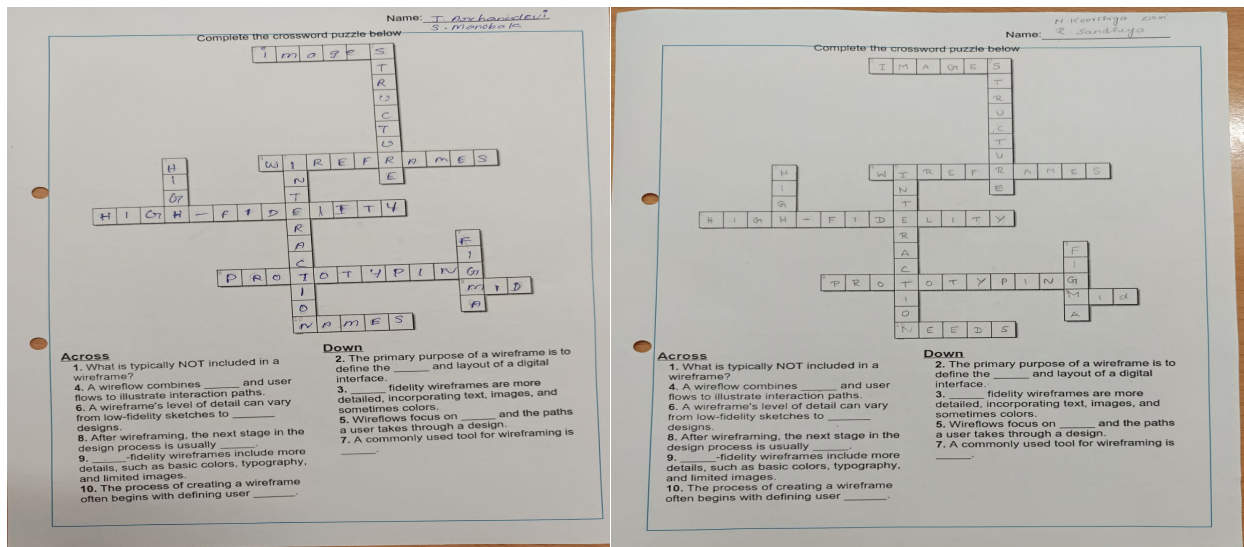


Fig 2: Sample sheets of solved crossword puzzle

• Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Students found the real-time simulation highly engaging and helpful in understanding complex concepts.
- Posting observations on the feedback wall helped peers learn different approaches to the algorithm.
- Students reported improved clarity in understanding the order of message recording and state capturing.

❖ Benefit of the practice:

- Enhanced conceptual understanding of the Snapshot Algorithm and FIFO channels.
- Encouraged active learning and teamwork.
- Students could visualize distributed system behaviour, making theoretical concepts tangible.
- Improved analytical and problem-solving skills related to message-passing algorithms

❖ Challenges faced in implementation:

- Some students initially struggled to track message order and channel states.
- Coordinating multiple teams in real-time simulation required careful facilitation.
- Time constraints made it challenging for all teams to complete the full simulation.

References:

- <https://www.ritrjpm.ac.in/images/computer-science/Crosswords-MCAP-GM.pdf>
- <http://www.classtools.net/crossword/>
- <https://www.activityvillage.co.uk/crosswords>